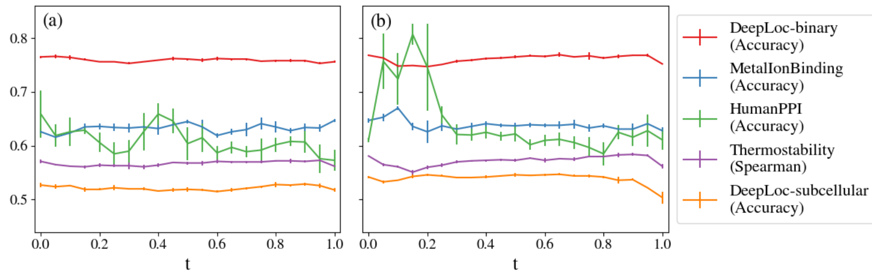

Figure 2: Evaluation of the t-dependence of the diffusion representation for the five protein property prediction tasks: **(a)** LSD-TN-M, **(b)** LSD-NM-M. The error bars are computed from the results of 5 randomly initialized predictors.

parameterised by t, but a definitive conclusion cannot be made. Again the HumanPPI metric stands out. We observe a consistent peak at $t = 0.15$ with value 0.807 ± 0.019, which (remarkably) is on a par with the ESM2 and DPLM 650M models. Given the inconsistent behaviour of the HumanPPI evaluation across all experiments however, it is difficult to conclude how meaningful this result is.

4.1 VISUALIZATION

To complement the above quantitative analysis, we provide a UMAP-based visual analysis of the learned representations in Fig. 3. We focus on the best performing LSD-NM diffusion representation, and use colouring to highlight the learned biological features. For each plot, we sample 64 sequences of length 100 amino acids from UniRef50, process them through the encoder and diffusion models, and employ UMAP to project the resulting embeddings to 2D.

We also conduct an attention map analysis to visualise how contextual information is integrated, and present this in appendix D.

5 DISCUSSION

The results of our evaluation highlight the key challenge in applying latent space diffusion to protein sequences: identifying an appropriate latent space. We observe that embeddings optimized for representation learning, e.g. those from the MLM baseline, result in an underperforming diffusion model. To address this, we proposed and analyzed alternative latent space learning methods designed to prioritise well-distributed embeddings. While these achieved the goal of boosting the diffusion model's performance, they ultimately fell short of matching the overall performance of token-based reconstructive learning methods like MLM, or the discrete diffusion method of DPLM.

Nevertheless, the autoencoder architectures we present have interesting features that may warrant further study. To our knowledge, the Token Norm bottleneck introduced here is novel. It is particularly suitable for protein sequence data, where the 20 amino acids provide a limited vocabulary compared to the much larger token vocabularies commonly used in NLP sequence modelling. The LSD-TN model is notable for its simplicity, achieving reasonable representation learning performance despite possessing a homogeneous bottleneck. We remark also that the univariate parametric form of the Kullback-Leibler divergence normalization loss is crude, and can perhaps be improved.

Our noise masking strategy for the LSD-NM model is quite similar to the diffusion denoising. A key difference however is that for the autoencoder the noise is applied inhomogeneously, while for the diffusion model it is applied uniformly across the sequence. The former places more emphasis on locality, while the latter learns the overall data distribution, underpining its generative capability. It may be interesting to explore if the two can be effectively combined. One possibility is to let the decoder and diffusion model share the same transformer trunk.

We comment also on the one-parameter representations offered by the t-dependence of the diffusion model. As described in Sec. 3 these amount to a regularised form of the score function, $s_t(z)$. We recall from the Introduction that the score function admits an interpretation as a distributional force.

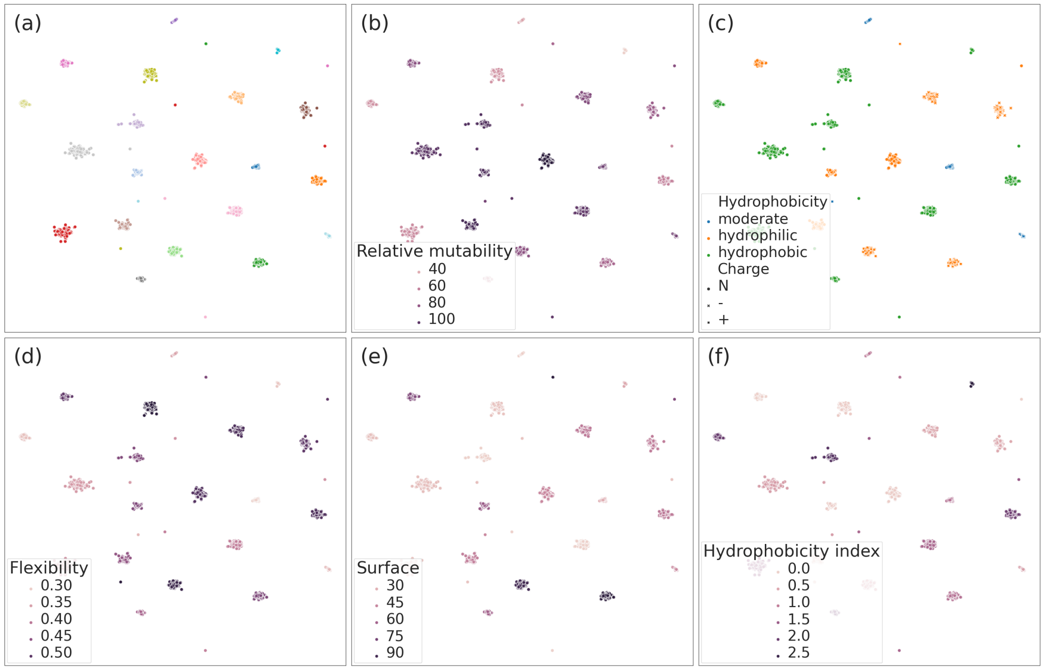

Figure 3: UMAP projections of the latent space learned by LSD-NM-M Diffusion model. **(a)** Coloured by amino acid. **(b)** Coloured by relative mutability Jones et al. (1992). **(c)** Coloured by hydrophobicity nature. **(d)** Coloured by hydrophobicity index Argos et al. (1982). **(e)** Coloured by average flexibility index Bhaskaran & Ponnuswamy (1988). **(f)** Coloured by residue accessible surface area in folded protein Chothia (1976).

Given that our models are trained on the evolutionary-scale Uniref50 dataset, we can thus offer an interpretation of $s_t(z)$ as a representation of the forces governing proteins, with the parameter t setting the scale of the latent space over which these forces are computed.

The DPLM representation included in Table 2 correspond to the $t = 0$ pass of their discrete diffusion model. It is unclear whether these can be extended non-zero t, as in this case the self-averaging property of Gaussian noise is lost, but this may be worthy of further investigation.

6 Conclusion and Outlook

We have presented a Latent Space Diffusion approach for modelling protein sequence data, with an initial focus on discriminative modelling. We highlighted the key challenge in developing this framework, which is to learn a sufficiently well-distributed latent space for the effective training of the diffusion model. To this end we proposed two novel autoencoder architectures: LSD-TN and LSD-NM. We evaluated their predictive performance across a range of protein prediction tasks and conducted an ablation study of key design choices. We found that while the diffusion performed better than with an MLM baseline, ultimately our trained models underperformed relative to token-based reconstructive learning approaches.

Our study provides an initial exploration of the LSD approach and opens up several interesting directions for future work. The architecture itself is not settled, and further research is needed to refine the question of what constitutes a good latent space. It remains unclear whether the inherent discreteness of sequence data can be sufficiently removed from the latent space to justify reconstructive learning through denoising. Another aspect is the latent space's dimensionality. It has been demonstrated that the learned embeddings of ESM2 can be massively compressed without significantly degrading their information content Lu et al. (2024b). This motivates a compression of the latent space, which in turn can facilitate more effective distributional modelling.

There is much to be learned about the richness of the information captured by the one-parameter family of diffusion representations, and how this can be best employed for protein modelling. Looking ahead, we also highlight that LSD could serve as a pre-trained model for fine-tuning on specific tasks. In particular, freezing the autoencoder while fine-tuning the diffusion model offers a particularly natural route forward, and may help to bypass the catastrophic forgetting issues observed in masked language models Wallat et al. (2021); Schmirler et al. (2024).

Another promising avenue is the incorporation of additional modalities, particularly protein structural data Mansoor et al. (2024). In this regard the continuous nature of the LSD formulation provides an advantage over discrete token-based approaches Hayes et al. (2024); Su et al. (2023).

Finally, it would be of great interest to scale up the model and explore its generative capabilities.

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